

ON TESTS OF EQUALITY OF TWO INVERSE GAUSSIAN DISTRIBUTIONS

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Summary: In this paper we consider the problem of testing the equality of two inverse Gaussian distributions on the basis of independent random samples from these distributions. We derive the likelihood ratio test and propose another test using Fisher's method of combining two independent test statistics. We derive the exact null distributions of these test statistics in simple form and give a numerical example. Finally, we show that both tests are asymptotically optimal in the sense of exact Bahadur efficiency.

1. INTRODUCTION AND THE PROPOSED TESTS

A random variable X is distributed as inverse Gaussian, denoted as $X \sim I(\mu, \lambda)$, if its density function is given by

$$f(x; \mu, \lambda) = \begin{cases} [\lambda/2\pi x^3]^{\frac{1}{2}} \exp[-\lambda(x-\mu)^2/2\mu^2 x], & x > 0 \\ 0, & \text{otherwise,} \end{cases}$$

where μ and λ are assumed to be positive. We have $E(X) = \mu$ and $\text{Var}(X) = \mu^3/\lambda$. As a failure-time model the inverse Gaussian distribution

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has recently received some attention. Folks and Chhikara (1978) have reviewed the statistical methodology of the inverse Gaussian distribution.

Fries and Bhattacharyya (1983) have listed some of the appealing features of this family of distributions:

- (a) a wide variety of shapes generated by the pdf makes it a prospective competitor to the gamma, Weibull and lognormal models;
- (b) it has the structure of an exponential family and the associated sampling distributions are well-known (see Tweedie (1957)); and
- (c) it is the distribution of the first hitting time of a positive barrier in a Brownian motion process with positive drift (see Cox and Miller (1965), page 221) and this provides a physical support to its applications in modelling waiting time data.

Suppose $X_{11}, X_{12}, \dots, X_{1n_1}$ is a random sample from $I(\mu_1, \lambda_1)$ and $X_{21}, X_{22}, \dots, X_{2n_2}$ is a random sample from $I(\mu_2, \lambda_2)$ and the two samples are independent of each other. We define $\theta = (\mu_1, \mu_2, \lambda_1, \lambda_2)$ and $\Omega = \{\theta; 0 < \mu_i < \infty, 0 < \lambda_i < \infty, i = 1, 2\}$. Let Ω_0 be a proper subset of Ω defined by $\Omega_0 = \{\theta; \mu_1 = \mu_2, \lambda_1 = \lambda_2\}$. The problem of testing the equality of μ_1 and μ_2 when $\lambda_1 = \lambda_2$ has been considered by Chhikara (1975), Miura (1978) and Davis (1980). In many situations the assumption that $\lambda_1 = \lambda_2$ may not hold. For example, in the context of experimental design the presence of unequal error variances due to different treatments has been recognized. "If the treatments consist of different amounts of lime applied to acid soil, the smallest dressings might give uniformly low yields with a small variance, while the highest dressings, being large enough to overcome the acidity, give good yields with a moderate variance. Intermediate dressings might give good yields on some plots and poor yields on others, and thus show the highest variance" (see Snedecor and Cochran (1967), page 324). This implies that a difference between two treatment means may be associated with a difference between the treatment variances and a null hypothesis of no treatment differences is equivalent to $\mu_1 = \mu_2$ and $\lambda_1 = \lambda_2$. We therefore wish to test the null hypothesis

$H_0: \theta \in \Omega_0$ versus the alternative hypothesis $H_a: \theta \in \Omega - \Omega_0$. A simple calculation shows that the likelihood ratio criterion V_n for testing H_0 versus H_a is given by

$$V_n = \prod_{i=1}^2 \left(\frac{n}{n_i} \right)^{n_i/2} \left(\frac{S_i}{S} \right)^{n_i/2}$$

where

$$S_i = \sum_{j=1}^{n_i} (x_{ij}^{-1} - \bar{x}_i^{-1}), \quad i=1,2$$

$$\bar{x}_i = n_i^{-1} \sum_{j=1}^{n_i} x_{ij}, \quad i=1,2$$

$$S = S_1 + S_2 + S_3,$$

$$S_3 = n_1 \bar{x}_1^{-1} + n_2 \bar{x}_2^{-1} - n^2 (n_1 \bar{x}_1 + n_2 \bar{x}_2)^{-1}$$

$$n = n_1 + n_2.$$

The random part of V_n is

$$\frac{\left(\frac{n_1}{2} \right)^{n_1/2} \left(\frac{n_2}{2} \right)^{n_2/2}}{\left(\frac{n}{2} \right)^{n/2}} \left(\frac{S_1}{S} \right)^{n_1/2} \left(\frac{S_2}{S} \right)^{n_2/2}.$$

We prove the following lemma.

Lemma 2.1 Under the null hypothesis H_0 , V_n is distributed as

$Z_1^{n_1/2} (1-Z_1)^{n_2/2} Z_2^{n_2/2}$ where Z_1 and Z_2 are independent beta random

variables with pdf's $\beta(z; \frac{n_1-1}{2}, \frac{n_2-1}{2})$ and $\beta(z; \frac{n-2}{2}, \frac{1}{2})$ respectively.

Proof: It is known that if $\mu_1 = \mu_2$ and $\lambda_1 = \lambda_2 = \lambda$, then the statistics λS_1 , λS_2 and λS_3 are independently distributed as chi-square random variables with n_1-1 , n_2-1 and 1 degrees of freedom respectively

(see Tweedie (1957) and Chhikara (1975)). If we let $U_{1n} = \frac{S_1}{S_1+S_2+S_3}$

and $U_{2n} = \frac{S_2}{S_1+S_2+S_3}$, then $U_n = U_{1n}^{n_1/2} U_{2n}^{n_2/2}$. The joint pdf of U_{1n} and

U_{2n} is

$$f(u_1, u_2) = \begin{cases} \frac{\Gamma(\frac{n-1}{2})}{\Gamma(\frac{n_1-1}{2}) \Gamma(\frac{n_2-1}{2}) \Gamma(\frac{1}{2})} u_1^{\frac{n_1-3}{2}} u_2^{\frac{n_2-3}{2}} (1-u_1-u_2)^{-\frac{1}{2}}, \\ \text{if } 0 \leq u_1 + u_2 \leq 1, 0 \leq u_i \leq 1, i=1,2; \\ 0, \quad \text{otherwise} \end{cases}$$

The h -th moment of U_n is

$$\begin{aligned} E[U_n^h] &= E\left[U_{1n}^{n_1 h/2} U_{2n}^{n_2 h/2}\right] \\ &= \frac{\Gamma(\frac{n-1}{2})}{\Gamma(\frac{n_1-1}{2}) \Gamma(\frac{n_2-1}{2}) \Gamma(\frac{1}{2})} \times B\left(\frac{n_1 h + n_1 - 1}{2}, \frac{1}{2}\right) \times B\left(\frac{n_2 h + n_2 - 1}{2}, \frac{n_1 h + n_1}{2}\right) \\ &= \left\{ \frac{\Gamma(\frac{n-2}{2}) \Gamma(\frac{n_1 h + n_1 - 1}{2}) \Gamma(\frac{n_2 h + n_2 - 1}{2})}{\Gamma(\frac{n_1-1}{2}) \Gamma(\frac{n_2-1}{2}) \Gamma(\frac{nh + n - 2}{2})} \times \left\{ \frac{\Gamma(\frac{n-1}{2}) \Gamma(\frac{nh + n - 2}{2}) \Gamma(\frac{1}{2})}{\Gamma(\frac{n-2}{2}) \Gamma(\frac{1}{2}) \Gamma(\frac{nh + n - 1}{2})} \right\} \right\} \\ &= E\left[Z_1^{n_1/2} (1-Z_1)^{n_2/2} Z_2^{n/2}\right]^h \end{aligned}$$

where Z_1 and Z_2 are independent beta random variables with pdf's

$\beta(z; \frac{n_1-1}{2}, \frac{n_2-1}{2})$ and $\beta(z; \frac{n-2}{2}, \frac{1}{2})$ respectively.

This proves the lemma.

From the lemma it is clear that the computation of the exact probability

$P_{H_0}(V_n \leq v)$, $0 < v < 1$, is very difficult. For practical purposes we can use the chi-squared approximation due to Box (1949). When H_0 is true $-2\rho \log V_n$ is approximately a chi-square random variable with 2 degrees of freedom where $\rho = 1 - \frac{1}{6}(\frac{1}{n_1} + \frac{1}{n_2}) - \frac{1}{12n}$. Hence, at an approximate level α , $0 < \alpha < 1$, the test rejects H_0 if and only if $-2\rho \log V_n > \chi^2(2; \alpha)$ where $\chi^2(m; \gamma)$ is the $100(1-\gamma)$ -th percentile of the chi-square distribution with m degrees of freedom.

Following Perng and Littell (1976) we describe another test of H_0 versus H_{1a} . We first consider the problem of testing the null hypothesis $H_{10}(\lambda_1 = \lambda_2)$ versus the alternative hypothesis $H_{1a}(\lambda_1 \neq \lambda_2)$. To test H_{10} versus H_{1a} , the likelihood ratio principle gives $F = F_{n_1, n_2} = (n_2 - 1)S_1 / (n_1 - 1)S_2$ df $n_1 - 1, n_2 - 1$ as a test statistic. We accept H_{10} at level α_1 if and only if

$$F(n_1 - 1, n_2 - 1, 1 - \frac{\alpha_1}{2}) < F < F(n_1 - 1, n_2 - 1, \frac{\alpha_1}{2})$$

where $F(m_1, m_2, \gamma)$ is the $100(1-\gamma)$ -th percentile of the central F -distribution with m_1 and m_2 degrees of freedom (see Davis (1980)).

Next we consider the problem of testing the null hypothesis $H_{20}(\mu_1 = \mu_2 | \lambda_1 = \lambda_2)$ versus the alternative hypothesis $H_{2a}(\mu_1 \neq \mu_2 | \lambda_1 = \lambda_2)$. To test H_{20} versus

H_{2a} we use $T = T_{n_1, n_2} = \{(n-2)S_3 / (S_1 + S_2)\}^{1/2}$ df $n-2$ as a test statistic. We accept H_{20} at level α_2 if and only if $-t(n-2, \frac{\alpha_2}{2}) < T < t(n-2, \frac{\alpha_2}{2})$ where $t(m, \gamma)$ is the $100(1-\gamma)$ -th percentile of the t -distribution with m degrees of freedom. It is known that the above tests are uniformly most powerful unbiased tests (see Chhikara (1975)).

We define the random variable W by

$$W = W_{n_1, n_2} = \begin{cases} 2[1 - G(F)], & \text{if } G(F) \geq \frac{1}{2} \\ 2G(F), & \text{if } G(F) < \frac{1}{2} \end{cases}$$

By the strong law of large numbers we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log K_n(X_n^*; \theta_1, \theta_0) = K(\theta_1; \theta_0)$$

with probability one for each $\theta_1 \in \Omega - \Omega_0$ where

$$\begin{aligned} 2K(\theta_1; \theta_0) = & -1 + \gamma \log \lambda_1 + (1-\gamma) \log \lambda_2 - \log \lambda \\ & + \lambda \gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) + \frac{\lambda \gamma}{\mu^2} (\mu_1 - 2\mu) \\ & + \lambda (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) + \frac{\lambda (1-\gamma)}{\mu^2} (\mu_2 - 2\mu). \end{aligned}$$

Let

$$J(\theta_1) = \inf \{ K(\theta_1; \theta_0); \theta_0 \in \Omega_0 \}.$$

A simple computation yields

$$\begin{aligned} 2J(\theta_1) = & \gamma \log \lambda_1 + (1-\gamma) \log \lambda_2 \\ & + \log \left[\gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) + (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) \right. \\ & \left. - \frac{1}{\gamma \mu_1 + (1-\gamma) \mu_2} \right]. \end{aligned}$$

We now proceed to show that the test based on Q_n is asymptotically optimal in the sense of exact slope. To find the exact slope of the sequence of test statistics $\{Q_n\}$ we need to compute the exact slopes of the sequences of test statistics $\{|T_{n_1, n_2}|\}$ and $\{W_{n_1, n_2}\}$. The exact slopes of $\{|T_{n_1, n_2}|\}$ and $\{W_{n_1, n_2}\}$ are given in the following lemma.

Lemma 3.1 The exact slopes of $\{|T_{n_1, n_2}|\}$ and $\{W_{n_1, n_2}\}$ are

$$\begin{aligned} c_1(\theta_1) = & \log \lambda_1 + \log \lambda_2 - \log \{ \gamma \lambda_2 + (1-\gamma) \lambda_1 \} \\ & + \log \left\{ \gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) + (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) \right. \\ & \left. - \frac{1}{\gamma \mu_1 + (1-\gamma) \mu_2} \right\} \end{aligned}$$

and

$$c_2(\theta_1) = \log \left\{ \frac{(1-\gamma)\lambda_1 + \gamma\lambda_2}{\lambda_1^{1-\gamma} \lambda_2^\gamma} \right\}$$

respectively, for each $\theta_1 \in \Omega - \Omega_0$.

Proof: From Example 5.1 on page 13 in Bahadur (1971), we have for $t > 0$

$$\begin{aligned} f(t) &= \lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) \log P_{H_0} [|T_{n_1, n_2}| > n^{\frac{1}{2}} t] \\ &= \lim_{n \rightarrow \infty} \left(-\frac{1}{n}\right) \log 2 P_{H_0} [T_{n_1, n_2} > n^{\frac{1}{2}} t] \\ &= \frac{1}{2} \log (1+t^2) \end{aligned}$$

and

$$b(\theta_1) = \lim_{n \rightarrow \infty} \frac{|T|}{n^{\frac{1}{2}}} = \frac{\left\{ \frac{\gamma}{\mu_1} + \frac{1-\gamma}{\mu_2} - \frac{1}{\gamma\mu_1 + (1-\gamma)\mu_2} \right\}^{\frac{1}{2}}}{\left\{ \frac{\gamma}{\lambda_1} + \frac{1-\gamma}{\lambda_2} \right\}^{\frac{1}{2}}}$$

with probability one for each $\theta_1 \in \Omega - \Omega_0$. Hence, using Theorem 7.2 on page 27 in Bahadur (1971), we get

$$\begin{aligned} c_1(\theta_1) &= 2f(b(\theta_1)) \\ &= \log \lambda_1 + \log \lambda_2 - \log \{ \gamma \lambda_2 + (1-\gamma) \lambda_1 \} \\ &\quad + \log \left\{ \gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) + (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) \right. \\ &\quad \left. - \frac{1}{\gamma\mu_1 + (1-\gamma)\mu_2} \right\} \end{aligned}$$

By definition, the exact slope of $\{W_{n_1, n_2}\}$ is

$$c_2(\theta_1) = \lim_{n \rightarrow \infty} (-2/n) \log W_{n_1, n_2}$$

with probability one for each $\theta_1 \in \Omega - \Omega_0$. When $\lambda_2 > \lambda_1$, we have with probability one

$$W_{n_1, n_2} = 2[1 - G(F)]$$

for all sufficiently large n . From Example 5.1 in Killeen, Hettmansperger, Sievers (1972) we have for $t > 1$

$$\lim_{n \rightarrow \infty} (1/n) \log\{1 - G(t)\} = \frac{1}{2} [\log \frac{t^\gamma}{(1-\gamma) + \gamma t}]$$

and $F = F_{n_1, n_2}$ converges to λ_2/λ_1 with probability one.

Thus, when $\lambda_2 > \lambda_1$ we have

$$\begin{aligned} c_2(\theta_1) &= \lim_{n \rightarrow \infty} (-2/n) \log\{1 - G(F)\} \\ &= \log \left\{ \frac{(1-\gamma)\lambda_1 + \gamma\lambda_2}{\lambda_1^{1-\gamma}\lambda_2^\gamma} \right\}. \end{aligned}$$

(The limit holds with probability one for each $\theta_1 \in \Omega - \Omega_0$.)

Similarly, when $\lambda_1 < \lambda_2$

$$c_2(\theta_1) = \log \left\{ \frac{(1-\gamma)\lambda_1 + \gamma\lambda_2}{\lambda_1^{1-\gamma}\lambda_2^\gamma} \right\}.$$

Lemma 3.2 The exact slope of $\{Q_n\}$ is

$$\begin{aligned} c(\theta_1) &= \gamma \log \lambda_1 + (1-\gamma) \log \lambda_2 + \log \left\{ \gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) \right. \\ &\quad \left. + (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) - \frac{1}{\gamma\mu_1 + (1-\gamma)\mu_2} \right\}. \end{aligned}$$

Proof: Using the results in Littell and Folks (1971) we have

$$\begin{aligned} c(\theta_1) &= c_1(\theta_1) + c_2(\theta_1) \\ &= \gamma \log \lambda_1 + (1-\gamma) \log \lambda_2 + \log \left\{ \gamma \left(\frac{1}{\mu_1} + \frac{1}{\lambda_1} \right) \right. \\ &\quad \left. + (1-\gamma) \left(\frac{1}{\mu_2} + \frac{1}{\lambda_2} \right) - \frac{1}{\gamma\mu_1 + (1-\gamma)\mu_2} \right\} \end{aligned}$$

for each $\theta_1 \in \Omega - \Omega_0$

We recall that V_n is the likelihood ratio criterion for testing H_0 versus H_a . Theorems 3.1 and 3.2 deal with the asymptotic optimality of the tests based on the statistics Q_n and $-(\frac{1}{n}) \log V_n$.

Theorem 3.1 For testing $H_0: \theta \in \Omega_0$ versus $H_a: \theta \in \Omega - \Omega_0$, the test based on Q_n is asymptotically optimal in the sense of exact Bahadur efficiency.

Proof: We note that $c(\theta_1) = 2J(\theta_1)$ for all $\theta_1 \in \Omega - \Omega_0$. The proof now follows from Corollary 3 in Bahadur and Ragahavachari (1972).

Theorem 3.2 For testing $H_0: \theta \in \Omega_0$ versus $H_a: \theta \in \Omega - \Omega_0$, the test based on $-(\frac{1}{n}) \log V_n$ is asymptotically optimal in the sense of exact Bahadur efficiency.

Proof: To prove the asymptotic optimality of the test based on $-(\frac{1}{n}) \log V_n$ we invoke Theorem 2.1 in Hsieh (1979). The two conditions for the applicability of this theorem are:

Condition A. $\liminf_{n \rightarrow \infty} \left\{ -(\frac{1}{n}) \log V_n \right\} \geq J(\theta_1)$ with probability one for each $\theta_1 \in \Omega - \Omega_0$.

Condition B. When H_0 is true, i.e. $\theta \in \Omega_0$, V_n is distributed as

$$M^{(n)} = [K^{(n)}]^{-1} \left[\prod_{i=1}^m Z_{1i}^{a_i^{(n)}} (1 - Z_{1i})^{b_i^{(n)}} \right] \prod_{j=1}^{m'} \frac{c_j^{(n)}}{Z_{2j}}$$

where Z_{ij} are independently distributed beta random variables such that $Z_{1i} (i=1,2,\dots,m)$ has density $\beta(z; a_i^{(n)} + a_i, b_i^{(n)} + b_i)$ and $Z_{2j} (j=1,2,\dots,m')$ has density $\beta(z; c_j^{(n)} + c_j, d_j)$; $m, m', a_i, b_i, c_j, d_j$ are fixed constants

independent of n ; $a_i^{(n)}, b_i^{(n)}, c_j^{(n)}$ are positive and depend on n such that as $n \rightarrow \infty$, $a_i^{(n)}/n \rightarrow \alpha_i$, $b_i^{(n)}/n \rightarrow \beta_i$, $c_j^{(n)}/n \rightarrow \delta_j$, $0 \leq \alpha_i \leq 1$, $0 \leq \beta_i \leq 1$, $0 \leq \delta_j \leq 1$; and

$$K^{(n)} = \prod_{i=1}^m \left[a_i^{(n)} / \{a_i^{(n)} + b_i^{(n)}\} \right]^{a_i^{(n)}} \left[b_i^{(n)} / \{a_i^{(n)} + b_i^{(n)}\} \right]^{b_i^{(n)}}$$

A straightforward computation shows that

$$\lim_{n \rightarrow \infty} \left\{ \left(-\frac{1}{n}\right) \log V_n \right\} = J(\theta_1)$$

with probability one for each $\theta_1 \in \Omega - \Omega_0$. Hence Condition A is satisfied.

Further, from Lemma 2.1 it follows that Condition B is also satisfied with

$m = 1$, $m' = 1$, $a_1 = -\frac{1}{2}$, $b_1 = -\frac{1}{2}$, $c_1 = -1$, $d_1 = \frac{1}{2}$, $a_1^{(n)} = \frac{n_1}{2}$, $b_1^{(n)} = \frac{n_2}{2}$, $c_1^{(n)} = \frac{n}{2}$, $\alpha_1 = \frac{\gamma}{2}$, $\beta_1 = \frac{1-\gamma}{2}$, $\delta_1 = \frac{1}{2}$. By Theorem 2.1 in Hsieh (1979) we conclude that $\left\{ \left(-\frac{1}{n}\right) \log V_n \right\}$ is an asymptotically optimal sequence for testing H_0 , with exact slope

$$c_3(\theta_1) = 2J(\theta_1) \quad \text{for all } \theta_1 \in \Omega - \Omega_0.$$

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